AI.450

Project

COURSE OVERVIEW

Capstone project: students design, build, and document an AI/robotics project from concept to demonstration.

PROJECT

1

SESSION

60 MINUTES

ORIENTATION: THE CAPSTONE

KEY TOPICS

- Project scope and deliverables
- Team formation
- Timeline and milestones
- Assessment criteria

TEACHING FOCUS

Plan a major AI/robotics project.

KEY TAKEAWAY A capstone synthesizes the degree.

PROJECT

2

SESSION

60 MINUTES

PROBLEM DEFINITION & REQUIREMENTS

KEY TOPICS

- Choosing a problem
- Stakeholder needs
- Functional and non-functional requirements
- Success criteria

TEACHING FOCUS

Define what the project must achieve.

KEY TAKEAWAY Clear requirements drive good projects.

PROJECT

3

SESSION

60 MINUTES

DESIGN & ARCHITECTURE

KEY TOPICS

- System design
- Hardware/software split
- Architecture diagrams
- Design review

TEACHING FOCUS

Architect the solution.

KEY TAKEAWAY Design before implementation.

PROJECT

4

SESSION

60 MINUTES

IMPLEMENTATION

KEY TOPICS

- Prototyping
- Iterative development
- Version control
- Documentation

TEACHING FOCUS

Build the system incrementally.

KEY TAKEAWAY Implementation turns design into reality.

PROJECT

5

SESSION

⏱ 60 MINUTES

TESTING & EVALUATION

KEY TOPICS

- Test plan
- Performance evaluation
- User testing
- Iteration

TEACHING FOCUS

Verify the system works.

▲ **KEY TAKEAWAY** Testing proves the project works.

PROJECT

6

SESSION

60 MINUTES

DEMONSTRATION & REPORT

KEY TOPICS

- Final demo
- Project report
- Presentation
- Reflection and future work

TEACHING FOCUS

Show and explain the completed project.

▲ **KEY TAKEAWAY** Communication completes engineering.